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PROJECT AND TEAM INFORMATION

Project Title

RakshaNetra - NHAI Worker Safety and Emergency Response System

Team Name: *Team-DSCPP-III-2026-T072*	
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PHASE-II PROJECT PROGRESS DESCRIPTION (35 pts)

Project Abstract

RakshaNetra aims to improve worker safety and emergency management at NHAI construction sites.It is difficult to manage worker information,verify identities,monitor site entry and exit and coordinate emergency when records are not together.
The main aim is to make a simple,organized system to manage worker and site details,record safety incidents,prioritizing emergencies and assign suitable rescue teams.The system maintains worker information such as identity,role,site and status.During an emergency the system checks the details, identify the affected worker,determine the severity and prioritize emergencies using a priority queue and then on the basis of emergency severity rescue teams are alloted.
The project uses Data Structures and Algorithms in C and Object Oriented Programming in C++.Arrays, linked lists, queues, searching and sorting records management and emergency prioritization. Classes,objects,inheritance and polymorphism help organize rescue team management.MySQL is used to store and access records.
The expected result is a working prototype that improves organization of worker information,supports severity based emergency prioritization and makes more efficient rescue team allocation. The future scope that we plan to add like real time tracking and automated alerts.

Updated Project Approach and Architecture

RakshaNetra manages worker safety and emergency response at NHAI construction sites.The system is divided in four parts worker and site management,emergency management and DSA,rescue team management using OOP and database integration and testing.
The system workflow is that first details of worker,construction site and rescue team are registered and stored.When emergency is reported the system checks the information,identifies the affected worker and finds the emergency severity. Priority queue is used to prioritize emergencies on the level of severity. The system identifies rescue team on the basis of requirement,equipment,location and availability and assigns the team to emergency.
The architecture uses C for data structures and algorithms and C++ for object oriented programming.Arrays,linked lists,queues,searching and sorting record management and processes emergency.Classes,objects,inheritance and polymorphism organize rescue team information.MySQL plans storage of worker,site,rescue team, emergency and assignment records in one place.Visual studio code is used to develop environment and Git and GitHub is used to support version control and collaboration.The C++ Standard Template Library support data structures such as priority queues.
The system is a prototype in which the components have defined functions and shared record identifiers such as Worker ID,Site ID,Emergency ID and Team ID. MySQL enable these parts to store and access records.Testing cover valid and invalid inputs,emergency prioritization,unavailable rescue teams and assignment of team successfully.
The current goal is to implement these parts in a working prototype which processes an emergency from reporting through rescue team allocation.

Tasks Completed

Task Completed	Team Member
Database for sites, workers, rescue teams,emergencies and incident history created	Pranjal Pandey
Severity level and priority queue insertion,display,processing and search logic implemented	Soumya Rawat
Linked list based incident history with add, display and search operations implemented	Sarthak Singh Chauhan
FIFO queue,stack,searching and bubble sort modules created and individual programs compiled and tested	Tazeen Anjum
Base rescue team and derived medical team,fire rescue team and general rescue team classes implemented and tested	Soumya Rawat
Sample emergency priority sorting tested. Allocation checks specialization and availability, general team backup and situations if no suitable team is available	Tazeen Anjum
x64 compiler and Connector/C++ include/library paths configured and databaseconnection.cpp compiled and linked to produce workflow	Pranjal Pandey

Challenges/Roadblocks

The first connector build used an x86 compiler against an x64 library causing linker errors. This was corrected by switching to the x64 Native Tools environment and the source now compiles and links.
VS Code reports that mysqlx/xdevapi.h cannot be found in its current IntelliSense configuration. This is separate from successful command line build and needs the include path in VS Code.
The connection executable currently exits without printing the expected success or failure message. The next step is to verify the executable or source version,connector runtime DLL availability, X Protocol port and service configuration and exception and output behaviour.
The C, C++ and MySQL are currently separate modules. They need to be still connected through shared identifiers and database read and write operations and then test cases will be integrated.

Tasks Pending

Task Pending	Team Member (to complete the task)
Resolve runtime MySQL connection and confirm successful connection output	Pranjal Pandey
Configure VS Code IntelliSense include paths for Connector/C++	Tazeen Anjum
Implement database reads and writes from C++ integration for workers, sites, emergencies, teams and incident history.	Soumya Rawat
Connect the modules in a single menu driven RakshaNetra interface.	Sarthak Chauhan
Add and complete worker and site management operations,input validation and record update and search flows.	Soumya Rawat
Run end to end tests for critical ,serious,moderate and minor incidents and busy teams and backup allocation,escalation,invalid inputs and history records.	Tazeen Anjum
Document setup and run instructions,capture screenshots and output evidence and prepare the final demonstration	Pranjal Pandey

Project Outcome/Deliverables

The main outcome is a functional prototype of NHAI Worker Safety and Emergency Response System which organize worker information and manages emergency.The deliverables include
Worker and Site Management Module in which registration,storage,searching,updating and management of worker and constructions its records including training and PPE status.
Emergency Management Module handles reporting emergencies, checking the details, deciding which emergency needs attention first based on its severity using a priority queue, and tracking its current status.
Rescue Team Management Module stores rescue team details and assigns available teams based on their skills, equipment, and location.
MySQL Database stores details of workers, sites, rescue teams, emergencies, and team assignments so that information can be managed properly and accessed easily.
Main Interface and Testing connects all these modules and tests different situations, such as deciding emergency priority, handling unavailable teams, checking incorrect inputs, and assigning teams successfully.
The project demonstrates the practical use of Data Structures and Algorithms in C and Object-Oriented Programming in C++. Its main goal is to organize records, help rescue teams respond to emergencies faster, and provide a foundation for future improvements like live location tracking and automatic alerts.

Progress Overview

We have almost completed sixty percent of the implementation as planned for the current phase.MySQL database structure has been created for storing worker details,rescue teams,accident sites,emergency severity and allocation records. The C part is developed to handle emergency situations using data structures and the C++ part shows object oriented programming concepts such as classes,inheritance, encapsulation and polymorphism.
The rescue team allocation part identifies an available team based on emergency type and it also checks for backup rescue teams when the required specialized team is not available and passes the emergency to the control room when no suitable team is available. Emergency seriousness and priority based sorting is implemented as the main idea.
The next phase is to integrate the C,C++ and MySQL parts in a complete one workflow in which we connect the allocation decisions with database records and test the complete system. We have planned some features as well for our future scopre.

Codebase Information

Repository Link: https://github.com/rawatsoumya23/Pblsem3
Branch: main
Project Folder: RAKSHANETRA
Main Folders: c_data_structures, cpp_oops, database, integration.
C Modules: emergency.c, incident_history.c, queue.c, searching.c, sorting.c, stack.c.
C++ Modules: rescue_team.cpp, allocation.cpp, EmergencyPriority.cpp, emergencyallocation.cpp.
Database Integration: database_connection.cpp and compiled test files.
Key Features: Emergency prioritization, incident history, rescue-team allocation, backup allocation, and escalation logic.

Testing and Validation Status

Test Type	Status	Notes
C modules compile/run individually	Pass (reported)	C modules were tested separately.
C++ rescue-team class demo	Pass (reported)	Team details and responses worked correctly.
Emergency priority sorting	Pass (reported)	Emergencies were sorted by priority. Priority 1 is the highest.
Team allocation / backup / escalation	Pass (reported)	Normal allocation worked. Backup needs more testing.
Connector/C++ compile and link	Pass	The program compiled without errors.
Runtime MySQL connection	Pending	Database connection is not yet confirmed.
Full database-to-allocation workflow	Pending	All modules are not yet connected into one application.

Deliverables Progress

MySQL database design has been completed which includes the structure for storing worker details, rescue teams,accident sites,emergency severity and allocation records.The C and C++ parts have also been made. The project shows data structures, classes,inheritance,encapsulation and polymorphism.The sorting of emergency based on severity and the rescue team allocation are completed in which available specialized team is selected and backup of rescue teams when there is none available.MySQL Connector/C++ has been installed and the connection program has been compiled but our successful database connectivity and data exchange are in progress. Complete integration of the C part ,C++ allocation and MySQL database and testing of different emergency scenarios is pending.Final project demonstration will be completed after system is integrated and tested.